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APPLICATION OF SOFT COMPUTING

A novel application of fuzzy inference system optimized with particle swarm optimization and genetic algorithm for PM_{10} prediction

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Abstract
Indoor air quality (IAQ) is a significant concern for occupant comfort, productivity levels, health, and well-being. The presence of harmful chemicals, inhalable particles, and toxic materials can be ten times worse as compared to outdoor air pollution. The increased concentration of PM_{10} pollutants in the indoor environment is a serious threat to human health in terms of critical respiratory health issues. Literature supports the application of artificial intelligence for the prediction of pollutant concentrations in building environments. In this paper, we proposed a novel application of fuzzy inference system (FIS) optimized using particle swarm optimization (PSO) and genetic algorithm (GA) for forecasting PM_{10} concentration. The study was conducted on experimental data obtained from an Internet of Things-based IAQ monitoring system. The real-time monitoring system measures four indoor air pollutants: PM_{10}, $PM_{2.5}$, CO_2, and VOC, along with two crucial thermal comfort parameters (Temperature and Humidity). The forecasting performance of the FIS with normalized dataset was measured in terms of MSE = 4.3656; MAE = 1.9351; MAPE = 9.633% and RMSE = 2.0894. Two optimization algorithms have improved RMSE FIS-PSO = 1.0746; and FIS-GA = 0.998. The optimized FIS can be used to develop real-time air quality prediction systems for indoor environments to promote health and well-being for building occupants.

Keywords Indoor air quality · Forecasting · Fuzzy inference system · Particle swarm optimization · Genetic algorithm · Public health

1 Introduction

Air quality is of significant interest in the scientific community over the last decades due to its impact on public health and well-being (Samet et al. 1987). Environment Protection Agency (EPA) has announced atmospheric pollution as a potential environmental problem on a global level (Kuklinska et al. 2015). The increasing pollutant concentrations destroy ecosystems, affect the atmosphere, introduce unrealistic climate change, disrupt photosynthesis, reduces crop harvests, and leaves a worse impact on biodiversity (Paoletti et al. 2010). The air quality in cities is greatly affected by traffic diversity, population density, degree of industrialization, meteorological variables, and topographical characteristics (Jun et al. 2014). The impact of air pollution is not limited to the outdoors. Moreover, air quality can be even ten times worse in the indoor environment (Kankaria et al. 2014). As human beings spend 80 to 90% of their routine time indoors, the impact can be more significant on their overall health and well-being (Saini et al. 2020a). Closed areas enable the fast build-up of pollutants in comparison with open spaces. As a result, indoor air pollution (IAP) outweighs the health impacts of outdoor pollution. IAP from harmful solid fuels leads to 4.5% global adjusted life year and 3.5 million deaths in the year 2010, out of which, 16% happened due to particulate matter (PM) pollution (Kankaria et al. 2014). Pollutant concentrations in the building premises are associated with chronic health problems, including respiratory illness (Armstrong and Campbell 1991), immunological deficiencies, reproductive problems,

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neurodegenerative diseases (Neurology 2018), malfunctioning hormonal systems and chronic obstructive pulmonary diseases (McCracken et al. 2012).

Principle sources contributing to IAP are bioaerosols, building materials, and combustion (Choo and Jalaludin 2014). In the developed nations, volatile organic matter, heavy metals, pesticides, asbestos, radon, and tobacco smoke are major indoor pollutants (European Environment Agency 2019). However, in developing countries, indoor pollutant concentration is closely associated with repeated usage of biomass fuel (Kankaria et al. 2014). In India, an estimated population of 200 million people, in rural areas, use biomass fuel for cooking and heating needs (Census of India Website 2020). Products of incomplete combustion of biomass fuels include poly organic matter, carbon monoxide, polyaromatic hydrocarbons, formaldehyde, and PM (Zhang and Smith 2003).

Repeated exposure to PM is more harmful to human health as compared to other pollutants (Ambient (outdoor) air pollution 2020). PM is the mixture of liquid droplets and solid particles hanging in the air (Ambient (outdoor) air pollution 2020). Furthermore, PM exists in three main forms: $PM_{0.1}$ (particles with diameter less than 0.1 μm), $PM_{2.5}$ (particles with diameters less than 2.5 μm), and PM_{10} (particles with diameter less than 10 μm). This study focuses on PM_{10} pollutants since it includes an inhalable fraction of liquid and solid particles that can directly reach the bronchi-tracheal area while leaving a considerable impact on lower airways (Brunelli et al. 2007). These particles can originate from several primary and secondary sources. Natural sources contributing to the generation of this harmful pollutant are living vegetations, forests, and grasslands, dust storms, marine salts, and volcanoes (Morawska et al. 2017). Moreover, human activities include domestic heating, vehicular traffic, construction works, industry, and central heating systems (Vautard et al. 2005). The chemical origin includes a mixture of inorganic and organic carbon, metals, phosphates, and nitrates (Ni et al. 2012). These particles can frequently travel into the indoor environment and become more hazardous in these closed spaces (Gumede and Savage 2017). Therefore, monitoring of these harmful air pollutants is a critical concern for occupant health (Babu and Suthar 2020).

In recent years, the potential emphasis is given to remote sensing applications for monitoring and detection of harmful air pollutants in the indoor environment (Saini et al. 2020b). However, most of these studies are either carried in the laboratory environment or have not provided reliable performance suitable for real-time environments (Saini et al. 2020b). The authors propose an indoor air quality (IAQ) monitoring and prediction system based on an effective combination of two emerging technologies: Internet of Things (IoT) and Artificial Intelligence (AI).

1.1 Related work

Several researchers have published studies on the use of genetic algorithm (GA), particle swarm optimization (PSO), Artificial Neural Network (ANN), and fuzzy logic controllers to design air quality prediction systems (Rahman et al. 2020; Li et al. 2022; Hadi and Abd 2021; Abdolrasol et al. 2021; Zhang et al. 2021). The study proposed in Lin et al. (2020) presents a neuro-fuzzy modeling approach to forecasting air pollutant concentrations. The main advantage of this system was that fuzzy elements could deal with the uncertainty of the impact factors, fuzzy clusters could handle the training data distribution, and fuzzy rules provided automatic training solutions for improved performance. The optimization algorithms PSO, GA, and gradient descent backpropagation in this study were used to optimize the input and output parameters for the fuzzy rule base to improve the learning process. The authors in Zhang et al. (2020) created a hybrid system using GA, ANN, fuzzy logic controller, and multivariate regression analysis (MRA) for optimizing building environment and energy consumption; however, it was not focused on forecasting the condition of particle pollutants. The observations state that GA helped to search for a reliable solution, while ANN helped to find the values of the individual objectives. MRA was further useful to reduce variable space. A fuzzy semi-supervised forecasting framework was designed by the authors of Bougoudis et al. (2018). It combined fuzzy logic, semi-supervised classification, and semi-supervised clustering to handle big datasets. A combination of ANN and GA was used by Antanasijević et al. (2013) for forecasting PM_{10}. The authors compared the performance with principal component regression and conventional multi-linear regression models. The results showed three times better performance. An optimal control strategy was proposed by Mossolly et al. (2009) for multi-zone air condition using GA. The proposed model presented reliable results in an experimental set up conducted by authors. A stepwise GA-based approach was developed by Mossolly et al. (2009). The proposed method provided optimized performance with modification of mutation and crossover operators. The authors in He et al. (2014) proposed a PM prediction system using ANN combined with a chaotic PSO algorithm. This optimized model presented improved prediction accuracy as compared to traditional algorithms such as PSO and Levenberg–Marquardt. In (Lu et al. 2002), the authors used ANN with PSO to analyze pollutant levels. It helped to overcome potential problems of conventional methods such as slow convergence and getting stuck at local minima. Another evolutionary technique for $PM_{2.5}$ forecasting was proposed by Bhardwaj and Pruthi (2020).

The proposed method combines the Wavelet Adaptive Neuro-Fuzzy Inference System (ANFIS) and PSO algorithm. The important findings of existing studies are highlighted in Table 1.

These studies suggest the potential of optimization algorithms to improve forecasting performance. To the best of our knowledge, none of the existing studies have used the combination of FIS adopting PSO, and GA optimization on air quality forecasting inside buildings. Therefore, the authors decided to test the efficiency of PSO and GA to enhance the performance of FIS for PM_{10} prediction. The FIS model was selected for its ability to deal with uncertainty in the large input data. Based on the previous studies, GA and PSO have been used for performance optimization. On the one hand, GA avoid local minima problems leading to improved accuracy. On the other hand, PSO has less computational cost. This analysis motivated us to test performance of different optimization techniques in combination with fuzzy systems to achieve desired forecasting results. The performance of both hybrid models was compared, and it was observed that the FIS-GA model outperforms the FIS-PSO model in terms of root-mean-square error (RMSE). This work proposed a novel application of FIS + PSO and FIS + GA with higher performance as compared to the existing conventional methods.

The remainder of this paper is distributed as follows. In Sect. 2, we provide details about the experimental design of IoT-based IAQ monitoring system and methods used for forecasting of PM_{10} concentrations. Section 3 presents the results of the proposed optimized model, and Sect. 4 includes a discussion on the results of the proposed model and performance in comparison with the existing models. Finally, the conclusion of this experimental analysis is presented in Sect. 5.

2 Materials and methods

Environment monitoring is an application of remote sensing in the area of IoT and wireless sensing networks (WSN) (Bhattacharya et al. 2012; Gomez et al. 2019). This section presents the development of an IAQ monitoring system based on an IoT sensor network. Moreover, the authors also describe the technical specification of the proposed prediction system.

2.1 Experimental design

The authors focused on measuring four indoor air pollutants: PM_{10}, $PM_{2.5}$, CO_2, and VOC, along with two main thermal comfort parameters (temperature and humidity). The sensor network was designed using three different sensor units where SDS011 was used to measure PM_{10} and $PM_{2.5}$, CCS811 for measuring CO_2 and VOC, and DHT11 sensor was used to measure temperature and humidity values in the indoor environment. Table 2 shows the specifications of used sensors for real-time measurements.

Table 1 Important highlights from existing studies

References	Used methods	Focus parameters	Remarks
Lin et al. (2020)	Neuro-fuzzy approach with PSO, GA and gradient descent method	PM_{10}, $PM_{2.5}$	Automatic extraction of rules, Correlation based input combinations
Zhang et al. (2020)	GA, ANN, MRA and fuzzy logic controller	Thermal comfort, indoor air quality, and energy cost	Optimal building design solution proposed for healthy environments
Bougoudis et al. (2018)	Fuzzy logic with semi-supervised classification and clustering	CO, NO, NO_2, O_3, SO_2	Multiple methods tested and compared, Massive amount of data used for analysis
Antanasijević et al. (2013)	ANN and GA	PM_{10}	Different methods were used for input selection for target parameter prediction
Mossolly et al. (2009)	GA	CO_2	Various control strategies were tested, Proposed optimization scheme helped to improved energy efficiency up to 30.4%
He et al. (2014)	ANN with chaotic PSO	PM_{10}	Seasonal variations on model performance were analyzed, Hybrid model was proposed for improving forecasting accuracy
Lu et al. (2002)	ANN, PSO	Respirable suspended particulates	Hourly and daily prediction performance was tested
Bhardwaj and Pruthi (2020)	Wavelet ANFIS, PSO	$PM_{2.5}$	Computational time was improved

Table 2 Manufacturer specifications for sensors

Sensor Name	Manufacturer	Measurement parameter	Typical range
DHT11	Aosong MPN	Temperature and humidity	0 °C to 50 °C; 20% to 90%
CCS811	SparkFun	CO_2 and VOC	0 ppb to 1187 pp b; 400 pp m to 8192 pp m
SDS011	Nova Fitness	PM_{10} and $PM_{2.5}$	0.0- 999.9 µg/m3

Although all these sensors come in factory calibrated form, authors have conducted separate reliability tests to ensure accuracy in the real-time measurements. The reliability test for DHT11 was conducted in the laboratory using Honeywell TM00X standard temperature and humidity meter. For the SDS011 laser sensor and CCS811 digital sensor, the reliability tests were conducted after 24 h of the preheating period, and the performance was measured against standard air monitoring station maintained and installed by Chandigarh air pollution board. The sensors showed stable performance within the range specified by the manufacturers.

The sensor network was installed in the cafeteria of the National Institute of Technical Teachers Training and Research, Chandigarh. The prototype was placed close to the cooking area to measure particle concentrations 24 h a day. The sensors were further connected to Arduino Uno microcontroller for real-time data measurement, and readings were collected at a gap of 15 min (four readings per hour). The data were further transferred to the ThingSpeak channel with the help of the ESP8266 wireless gateway module. The real-time monitoring for six IAQ parameters was done for the total duration of six months from July 10th, 2019 to January 9th, 2020. The general architecture of IoT based monitoring system is given in Fig. 1.

2.2 Data preprocessing

Real-time monitoring of IAQ parameters in the target location was affected by several conditions. These include unusual system failures and sudden power shutdowns. This scenario caused some missing values in the dataset that are to be handled before using the recorded data for the proposed forecasting system design. We used the mean imputation method in MATLAB R2020a environment where all the missing values were first marked to be zero. Then, the respective zeros were replaced with a calculated mean value of the respective IAQ parameter as recommended by Sim et al. (2015); Wijesekara et al. 2020). This initial data cleaning resulted in a total of 17,280 samples for measured six IAQ parameters. Before designing the PM_{10} forecasting system, the authors analyzed the statistical description of recorded real-time data using the IBM SPSS software. Table 3 provides a statistical description of all six measured parameters in terms of min, max, mean, median, and standard deviation.

Statistical description of the data shows that PM_{10} and $PM_{2.5}$ values are highly correlated. The cross-verification for the same was conducted with the help of Spearman correlation, where $PM_{2.5}$ showed a positive correlation value of + 0.997 concerning the target variable PM_{10}. However, it was + 0.307, − 0.280, − 0.337, and − 0.285 for VOC, CO_2, temperature, and humidity parameters, respectively. Therefore, to design the PM_{10} forecasting system, the $PM_{2.5}$ variable may cause an overfitting problem, which can further generate misleading conclusions (Fennell et al. 2019). Also, the literature shows that the performance of the forecasting model is greatly influenced by feature selection (Kim et al. 2009). It leaves a considerable impact in terms of error performance, time-taken to produce the best performance by the model, and for real-world implementations. Consequently, the authors discarded the $PM_{2.5}$ parameter at the model training stage.

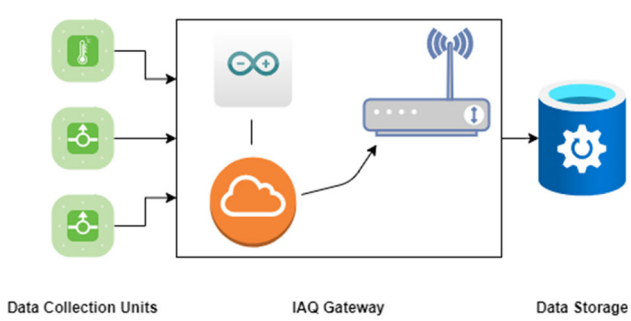

Fig. 1 General architecture of IAQ monitoring system

Table 3 Statistical description of the dataset

Parameters	Min	Max	Mean	Median	STD
PM10	0.5	89.20	14.3	9.5	11.95
PM2.5	0.5	85.20	13.3	8.7	11.46
CO2	381	2626	833.6	641	475.8
VOC	0.062	0.59	0.097	0.065	0.067
Temperature	19.42	27.96	23.69	23.52	11.5
Humidity	31.44	72.09	42.17	41.21	4.52

Four parameters (VOC, CO_2, temperature, humidity) were used as input features for the training forecasting model with PM_{10} as the target variable.

Air quality time series data with independent variables can have wide range, and all parameters measured in this process have different units. Direct analysis of such data may cause variability and overfitting problems (Grange and Carslaw 2019). Therefore, it is necessary to normalize this large-scale data before training to ensure enhanced accuracy (Cai et al. 2020).

In this study, the authors have used a z-score normalization on input features of the dataset as recommended by Srivastava et al. (2018), Aggarwal and Toshniwal (2019). This method was applied in MATLAB environment. This normalization technique returns vector-wise z-score of data with standard deviation one and center 0. The normalization function used for this study is described by Eq. 1 (Srivastava et al. 2018).

$$z = \left(x - \overline{X}\right)/S \tag{1}$$

where x is the original data before normalization, z is the z score normalization of the dataset. $\overline{X}$ represents the mean of the input data, and S stands for standard deviation.

The normalized dataset was further separated into training and test set. We use a random sampling method for validation. 70% of data was used for training, and the remaining 30% was used for model testing. We preferred random sampling to avoid biasing problems in model training as suggested by Suresh et al. (2011), Xu and Goodacre (2018). The model performance was tested on normalized dataset, as well as on dataset without normalization.

2.3 Fuzzy inference system, particle swarm optimization and genetic algorithm

In recent years, AI offered great scope for the design of real-time prediction systems for remote sensing applications. The advanced algorithms in the field of machine learning, neural network, fuzzy logic, and optimization promise enhanced decision-making for intelligent prediction systems. The authors used FIS in combination with PSO and GA to predict PM_{10} in indoor environments.

2.3.1 Fuzzy inference system

Fuzzy inference is an advanced approach in fuzzy logic that interprets the values in the input vector depending upon a set of pre-defined rules and assigns respective values to the output or target vector. The process of fuzzy inference involves various aspects, including fuzzy logic operations, membership functions, and if–then rules. Two widely used FIS are Mamdani FIS and Sugeno FIS (Kalogirou and Kalogirou 2009). Mamdani type FIS expects the output membership functions to be fuzzy sets. The completion of the aggregation process leads to a fuzzy set for every output variable, which is further de-fuzzified to get desired results (Fan et al. 2019). The generalized structure of the FIS model is given in Fig. 2.

In the fuzzification process, a specific set of values enters the FIS, and they are converted to a membership level based on the pre-defined rules. The rule base consists of the actual design methodology for FIS in the form of linguistic variables (If–Then rules). The rule base or knowledge base mainly has two parts: antecedent and consequent; both are represented by membership functions described as per rule base. The output corresponds to a fuzzy set, and it is further de-fuzzified to get the desired numerical data. The commonly used de-fuzzification methods are centers average and centroid.

2.3.2 Optimization algorithms

GA and PSO are AI-based optimization techniques that imitate living systems to find optimized solutions for real-world problems (Jatana and Suri 2020). The literature shows extensive use of GA for solving computational problems, and PSO is widely used for the continuous search space (Kennedy and Eberhart 1995; Hassan et al. 2005). Consequently, we used two meta-heuristic algorithms to optimize the performance of the Mamdani FIS system.

PSO was initially developed by Eberhart and Kennedy in the year 1995 (Bhardwaj and Pruthi 2020; Kennedy and Eberhart 1995). This algorithm is influenced by the bird flocking, and this algorithm is considered an intelligent solution in terms of implementation and architecture. Moreover, the PSO algorithm makes use of a lesser number of parameters in comparison with other famous evolutionary methods (Jatana and Suri 2020). Two critical variables are updated after every iteration: particle position and particle velocity. For every iteration, the performance is optimized based on the current global best value, the present best value, inertia weight, and acceleration. The output values are further used to update the particle

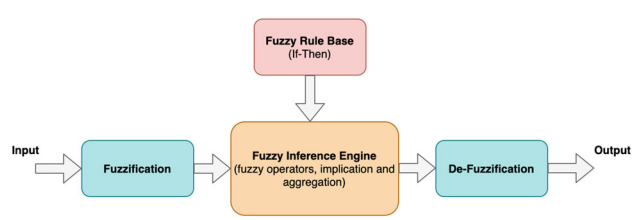

Fig. 2 General FIS Structure

velocity and position in Eqs. 2 and 3 (Jaberipour et al. 2011).

$$v_i^k = wv_i^k + c_1 r_1 \left(pbest_i^k - x_i^k\right) + c_2 r_2 \left(gbest_i^k - x_i^k\right) \quad (2)$$

$$x_i^{k+1} = x_i^k + v_i^{k+1} \quad (3)$$

where v_i^k is the velocity of the ith particle during kth iteration, x_i^k is the current position (solution) of the ith particle during the kth iteration. r_1 and r_2 and random variables having uniform distribution between 0 and 1; whereas c_1 and c_2 are acceleration constants. Furthermore, w in this equation represents inertia weight showing the impact of the previous velocity value on the new vector (Jaberipour et al. 2011).

GA was first introduced by John Holland in the year 1970. It has been used to find solutions for various types of problems over the years. Furthermore, GA follows the concept of evolutionary biology (Bhardwaj and Pruthi 2020). The structure is further based on the idea of mutation, crossover, and inheritance. The crucial feature of GA is that it performs efficiently, even with a large number of variables, and shows better performance as compared to other evolutionary techniques (Yuan et al. 2014). The operations are performed based on a pre-declared population of chromosomes instead of a single chromosome. The performance of the algorithm is measured in terms of a fitness function or objective function, same as PSO (Goldberg and Holland 1988).

2.3.3 Proposed prediction system

Mamdani FIS system was used by several works available in the state of the art (Ganesh et al. 2017; Javid et al. 2016). Consequently, we used the Mamdani FIS system for designing the proposed prediction system with four input variables (CO_2, VOC, temperature, humidity) and one target/output variable (PM_{10}). The performance of the FIS model was further enhanced with the help of PSO and GA, which helped to reduce system RMSE by a considerable level. The overall methodology of the proposed system is given in Fig. 3. The recorded IAQ and thermal comfort data are first transferred to the preprocessing unit that works in four steps. First, we use the data cleaning approach by implementing imputation methods. Second, the statistical analysis was applied to the processed data. The statistical description provided correlation details for all features corresponding to the target variable. Concerning the correlation information, the $PM_{2.5}$ feature is rejected, and the rest four variables (CO_2, VOC, temperature, and humidity) are accepted as input parameters for model training. Finally, the selected features are normalized, and the data are further transferred to the input stage of the FIS model.

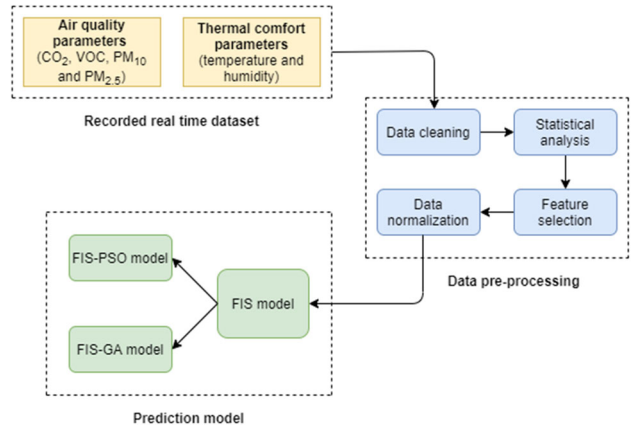

Fig. 3 Methodology of the proposed prediction system

The FIS model was further trained using a set of rules that establish a link between input IAQ parameter concentrations and target parameter concentration levels. The model performance was evaluated in terms of mean square error (MSE), mean absolute error (MAE), RMSE, and mean absolute percentage error (MAPE). The proposed prediction model was optimized using PSO and GA algorithms to improve performance.

Normalized data after feature selection were used for designing the FIS-based prediction model. The proposed architecture of FIS is based on the combination of selected four inputs and one target variable.

EPA has already defined pollutant threshold levels that divide concentrations into different categories, such as good, moderate, unhealthy, and hazardous (Dionova et al. 2020). The proposed Mamdani system uses three different levels for input pollutant concentrations as good, moderate, and poor, which are defined with the help of triangular membership functions. Figure 4 shows membership functions for all four normalized input variables.

For input membership functions, the FIS system assesses concentrations of the IAQ parameters based on z-score

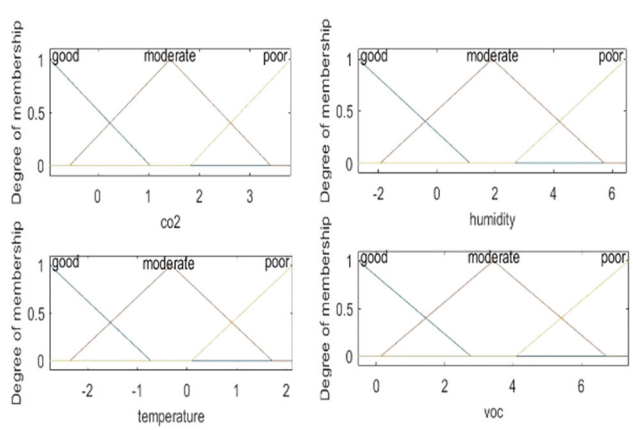

Fig. 4 FIS membership functions for four input variables

normalization, utilizing the threshold levels represented in Table 4 (Nihalani et al. 2020; Saddek et al. 2014).

The output variable is assigned six different levels (good, satisfactory, moderate, poor, very unhealthy, severe) using six triangular membership functions to represent each level. A large number of system parameters with a higher number of membership functions lead to more multiplications causing computational complexity for the FIS network (Talpur et al. 2017). Therefore, the authors preferred using a limited number of membership functions at the input and output side (Wongsathan 2018). Table 5 represents the threshold levels and respective ranges for the target variable (PM_{10}). This technique helps to reduce convergence time and improve network performance.

After assigning membership functions to the fuzzy parameters, the rule base for FIS can be represented. The number of rules to be used in the FIS knowledge base relies on the potential membership function combinations at input and output levels. The rule base for the proposed system was constructed by utilizing individual combinations of input air quality parameters, which further helps to define the condition of the output variable. The proposed system includes 48 rules that consist of good, moderate, and poor conditions at the input side and good, satisfactory, moderate, poor, very unhealthy, and severe levels at the output side. Each rule base utilizes all four input variables, one output variable, and one weight vector written in () by the end of every rule. For a better understanding of the rule base, one sample rule is given as follows:

$$\text{“}co2 == good \ \ \& \ humidity == good \ \ \& \ temperature = \\ = good \ \ \& \ voc == good = \ > pm10 = good(1)\text{”}$$

The complete list of implemented rules is presented as supplementary material (Appendix A).

3 Results

The conditions of the particle pollutant concentrations are clearly defined. Therefore, the proposed FIS is capable of identifying the potential emergency. After completing the development of the knowledge base for FIS, the final task is to do the conversion of fuzzy output values into defuzzified values or crisp values. Consequently, the proposed FIS prediction system makes use of the centroid method to determine the PM_{10} concentration level. The design of the proposed FIS is shown in Fig. 5. It shows four inputs, each having three membership functions, one output having six membership functions, and the fuzzy rule bases having 48 number of rules for Mamdani type FIS.

Table 5 Threshold points for the output variable

PM_{10} (μg/m^3)	Status
0–50	Good
40–100	Satisfactory
90–250	Moderate
240–350	Poor
340–400	Very Unhealthy
390–500	Severe

The performance of the proposed system was measured using four metrics: MSE, MAE, MAPE, and RMSE. On the one hand, the proposed PM_{10} forecasting system trained using normalized data with FIS model gives MSE = 4.3656; MAE = 1.9351; MAPE = 9.633% and RMSE = 2.0894 on the test dataset. On the other hand, the performance of FIS model was measured as MSE = 259.46; MAE = 12.97; MAPE = 1.875%; RMSE = 16.10 when trained without normalizing dataset. The model performance was further improved with the help of PSO and GA, which further helped to reduce the RMSE of the forecasting system by a considerable level. The comparative performance of FIS, FIS-PSO, and FIS-GA models is given in Table 6.

The FIS-PSO and FIS-GA model performance for normalized and not normalized dataset is shown in Figs. 6, 7, 8 and 9, respectively.

The results show that forecasting performance was considerably improved by using the optimization algorithm. However, the GA algorithm outperformed PSO. The main advantage of the PSO algorithm over GA was that it consumed lesser time (4.52 min) to give the best solution with normalized dataset. However, GA took 22.34 min to achieve its best performance. The proposed hybrid models can be used for real-time IAQ monitoring and prediction in the indoor environment.

Table 4 Threshold points for input variables

CO_2 (ppm)	VOC (ppm)	Temp (°C)	Hum (%)	Status
0–606	0–0.1	18–25	40–70	Good
580–1020	0.08–0.3	22–29	60–80	Moderate
800–5000	0.26–3	26–45	70–100	Poor

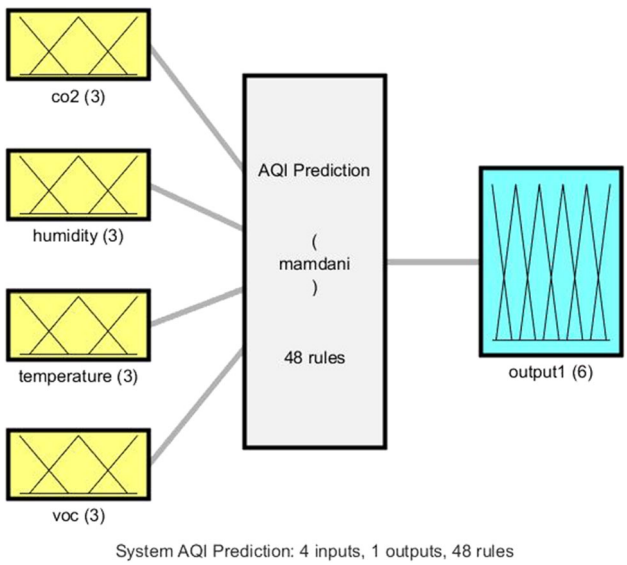

Fig. 5 Structure of the proposed FIS model for PM10 forecasting

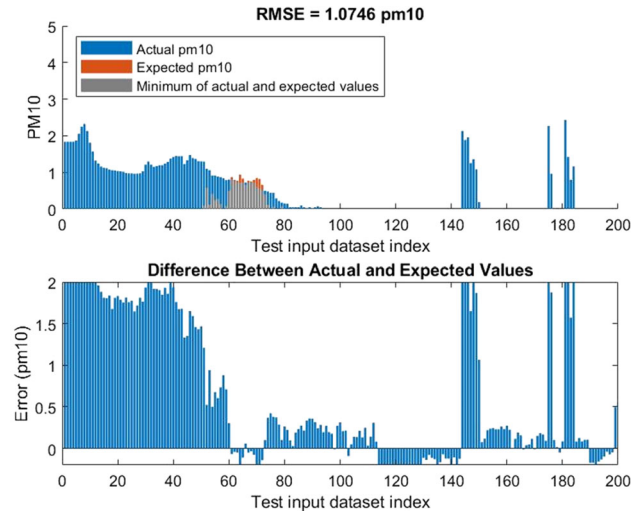

Fig. 6 FIS-PSO performance in terms of RMSE for normalized dataset

4 Discussions

The detailed description of the development of the IAQ forecasting model is explained in detail in the above sections. As described in Sect. 2, the real-time data for designing this prediction model were obtained from a self-designed IoT-based monitoring system. The monitoring system helped to measure six important IAQ parameters: PM_{10}, $PM_{2.5}$, CO_2, VOC, temperature, and humidity. The authors also worked on correlation analysis of explanatory (input) variables concerning the target variable (PM_{10}). The features for model training were decided based on the correlation information. A normalized dataset was prepared for designing the FIS model for PM_{10} forecasting. In Sect. 1, it was discussed that IAP is a potential cause of several potential health problems. Therefore, the forecasting of pollutant concentrations is a critical concern for public health. It is crucial to design a reliable, cost-effective, and efficient model for making predictions regarding future conditions. The main goal behind forecasting is to guide building occupants about critical threshold levels of pollutant concentrations. Consequently, the building manager can take appropriate ventilation measures ahead of

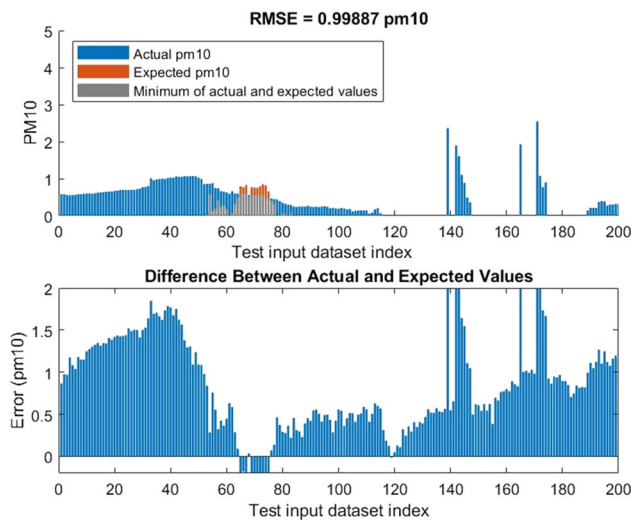

Fig. 7 FIS-GA performance in terms of RMSE for normalized datset

Table 6 Performance comparison for FIS, FIS-PSO, and FIS-GA in terms of RMSE

Model	Number of iterations/generations	Population size	Time taken (min)	RMSE
FIS (Normalized dataset)	–	–	0.30	2.0894
FIS	–	–	0.45	16.10
FIS-PSO (Normalized dataset)	20	50	4.52	1.0746
FIS-PSO	50	50	10.55	11.93
FIS-GA (Normalized dataset)	20	50	22.34	0.998
FIS-GA	50	50	38.56	10.483

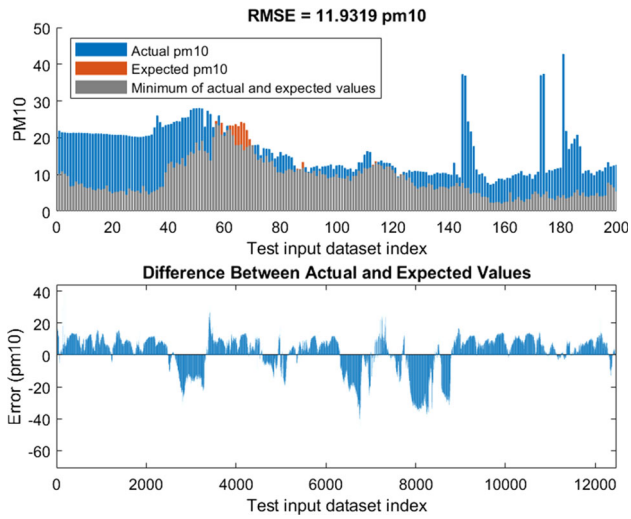

Fig. 8 FIS-PSO performance in terms of RMSE without normalization

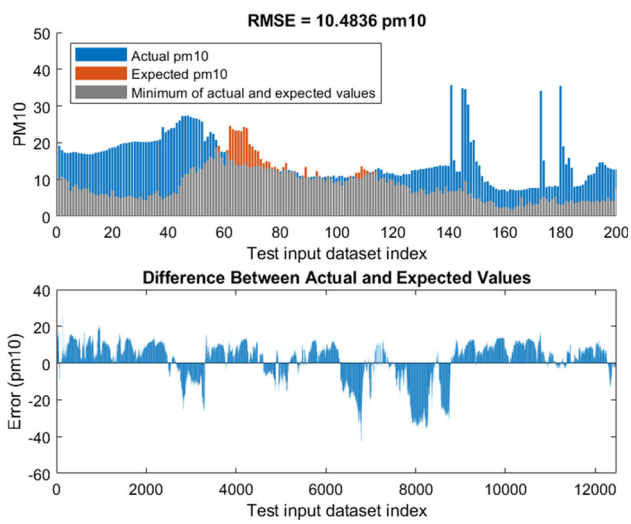

Fig. 9 FIS-GA performance in terms of RMSE without normalization

time. As human beings spend 80 to 90% of their routine time indoors, either in the home or office, staying aware of poor pollutant concentration levels could be a great initiative to prevent critical health consequences (Saini et al. 2020a).

The non-optimized FIS model trained with normalized dataset was not that efficient to work in real-time conditions with RMSE = 2.0894. However, it was improved by a considerable level by applying optimization algorithms. PSO and GO optimization algorithms have unique advantages and disadvantages. These are studied by existing researchers over the years (Sharma and Singhal 2015; Wihartiko et al. 2018).

The computational cost of PSO is generally lower than GA. This is because the number of function evaluations in the case of PSO is considerably lesser than GA (Wihartiko et al. 2018). Moreover, PSO is faster as compared to GA in terms of conversion. However, when tuned adequately, GA is efficient enough to avoid local minima problems (Nobile et al. 2018). Other than this, the crossover, mutation, and selection processes in GA give it the ability to handle multi-object problems more efficiently. The FIS-GA algorithm used in this study provided comparatively better performance, which makes it suitable for real-time applications. The model performance was also compared with other existing models in the literature. Table 7 provides comparative details of the proposed system with existing studies in the literature.

The proposed models present better performance than the existing studies in the literature. Particularly, the proposed FIS-GA model with normalized dataset outperformed all with better RMSE performance. This model provides PM_{10} forecasts based on the data collected from the cafeteria of an institute in Chandigarh city, India. The model was trained over 70% training data, and then, performance was analyzed on the remaining 30% test data. The results indicate that RMSE values for the FIS-GA model are close to zero in comparison with other models.

There have been numerous research studies published on different methods used for forecasting IAQ pollutant concentrations (Bai et al. 2018). Studies report that ARIMA models are efficient to conduct forecasting tasks, whereas others consider ANFIS as a more reliable solution for time series forecasting applications (Yadav and Balakrishnan 2014). It is also possible to find a variety of advantages and disadvantages of these models over each other. However, we must note that the results of specific models for forecasting applications vary depending upon a few essential factors. These factors include the type of dataset, scale or range of numerical inputs, the variability of parameters and units, feature selection, normalization/standardization techniques, and model parameters at the time of training (Qian et al. 2019; Graefe et al. 2015; Wang et al. 2016).

Furthermore, when the forecasting system is desired to be installed in real-world scenarios, it is also crucial to analyze some output related aspects such as computational complexity, resource utilization, time taken by a model for generating the best performance, and value of performance metrics.

Authors in Wongsathan (2018) used the ANFIS model on normalized IAQ dataset since it is known for its property of having both linguistic and numerical knowledge. They used ANN and the FIS model's capability to deal with uncertainty in data (Ludwig 2019). However, the main disadvantage of ANFIS is that it is computationally expensive and complex as well (Yadav and Balakrishnan 2014). Therefore, we preferred the FIS method instead of

Table 7 Comparison of the proposed forecasting model with existing studies

References	Method	RMSE
Wongsathan (2018)	ANFIS model (normalized dataset)	RMSE = 7.3
Yang et al. (2020)	Convolution Neural Network-Gated Recurrent Unit (CNN-GRU) (normalized dataset)	RMSE = 1.688
Kim et al. (2009)	Recurrent Neural Network	Case 1:
		PM10: RMSE = 29.37
		Case 2:
		PM10: RMSE = 28.57
Hamid et al. (2016)	ARIMA model	RMSE = 11.992
Xayasouk et al. (2020)	Long Short-Term Memory Network (LSTM) and Deep Autoencoder (DAE)	LSTM Model:
		RMSE = 11.113
		DAE Model:
		RMSE = 15.038
Lin et al. (2020)	Neuro-fuzzy network, PSO, GA, and gradient descent backpropagation methods	Station 1:
		RMSE = 10.92
		Station 2:
		RMSE = 13.24
		Station 3:
		RMSE = 11.84
		Station 4:
		RMSE = 15.58
Antanasijević et al. (2013)	General regression neural network (GRNN) and GA (normalized dataset)	RMSE = 3.96
Proposed	FIS model	RMSE = 16.10
	FIS model (normalized dataset)	RMSE = 2.0894
Proposed	FIS-PSO model	RMSE = 11.93
	FIS-PSO model (normalized dataset)	RMSE = 1.0746
Proposed	FIS-GA model	RMSE = 10.4836
	FIS-GA model (normalized dataset)	RMSE = 0.998

ANFIS to propose a computationally efficient model that can be developed for real-world scenarios.

The CNN-GRU model used in Yang et al. (2020) also presented a considerable performance for predicting PM concentrations with normalized dataset. The main disadvantage of GRU is low learning efficiency and slow convergence. The work proposed by the authors of Yang et al. (2020) designed an optimized model to achieve enhanced performance with CNN. However, the model was extensive with 64 neurons at the one-dimensional convolution layer, and 32 GRUs at the second recurrent layer.

Recurrent neural network (RNN) was proposed by the authors of Kim et al. (2009). This method is a type of neural network that use the learnings from previous stages to forecast future trends. The main challenge with these networks is that they can remember only a few steps within the input data sequence. Therefore, RNN does not provide efficient results for longer sequences. This can be seen from the performance of Kim et al. (2009) with a higher RMSE value (28.57) on test data. However, one advantage of this study is that authors utilized input data in two sets. The first dataset contained all the recorded variables (Case 1) and the second dataset contained only three sensitive parameters based on the relation between dependent and independent variables (Case 2). The model presented considerable improvement when trained with a selected number of features instead of when trained with all set of features. We have followed a similar approach to Kim et al. (2009). The proposed FIS model was also tested for such two cases to analyze the performance. When trained with five input features including highly correlated $PM_{2.5}$ along with four other least correlated features CO_2, VOC, temperature, and humidity, the RMSE of the model was 7.53. However, it was reduced up to 2.0894 when trained with a selected number of features (CO_2, VOC, temperature, and humidity) based on the statistical description.

The authors in Hamid et al. (2016) used the ARIMA model for pollutant concentration forecasting, and it is widely accepted for its ability to handle time-series data. The authors in this paper worked on yearlong data from

April 2008 to March 2009 while considering all seasonal changes. The study also provides details about statistical features of the measured PM_{10} pollutant. However, the system performance in terms of RMSE (11.992) was not that efficient for real-time implementation.

LSTM neural network proposed in Xayasouk et al. (2020) has a particular ability to memorize a longer sequence of data with the help of gates. Nevertheless, the main limitation of these networks is that they require an extensive range of resources. Consequently, LSTM requires more time to complete training and is not suitable for real-world applications. The authors of Xayasouk et al. (2020) show the estimated processing time of 38.18 min to generate the best performance for PM_{10}. The processing time is higher than the proposed model in this study that is 22.34 min for FIS-GA and 4.52 min for FIS-PSO. We have tested three different techniques for forecasting of PM_{10} concentrations in the indoor environment. The FIS model includes 48 rules based on the input and output membership functions. The proposed model shows a RMSE value of 2.0894 using four normalized input features. Furthermore, we have utilized PSO and GA for enhanced system performance. The authors observed that on application of PSO, the RMSE over normalized dataset was reduced up to 1.7, whereas when optimized with GA, the RMSE was improved up to 0.998. However, the time required by FIS-PSO algorithm was 4.52 min whereas for FIS-GA was 22.34 min. Consequently, we recommend the utilization of FIS-GA in systems with higher computational resources and the use of FIS-PSO in systems with limited computational resources.

The methods used for (Lin et al. 2020) and (Antanasijević et al. 2013) have been discussed in the introduction section. Lin Y-C et al. (Lin et al. 2020) used PSO and GA to optimize the neuro-fuzzy network along with gradient descent backpropagation algorithm to create a hybrid learning model. The results show that performance after optimization was considerably improved as compared to the conventional methods. However, the best RMSE of the combined model was 4.8255 only. Moreover, the authors of Antanasijević et al. (2013) used the GRNN model where GA was used for input variable selection. GA helped to eliminate less significant features to improve forecasting performance over conventional methods. The results show RMSE achieved by the system was 3.96. Authors in Antanasijević et al. (2013) preferred using normalized dataset to improve the model performance.

The proposed system in this study presented enhanced performance as compared to other studies listed in Table 6. Nevertheless, no system can be proposed without limitations. The main limitation of the proposed forecasting model is that it makes use of a six-month dataset and six IAQ parameters. Other relevant pollutants such as ozone, sulfur dioxide, carbon monoxide, nitrogen monoxide that leave a considerable impact on human health in indoor environments are not considered in this study. Therefore, the authors are planning to consider them for future work. Another challenge for future work is to deal with the trade-off between performance and time taken by the model for forecasting. We can observe that the time taken by PSO to give the best solution was only 4.52 s, whereas it was 22.34 min in the case of GA-based optimization (Table 5). The real-time applications of forecasting systems require fast response without compromising accuracy. For this, the authors are planning to work on varying initial parameters of optimization algorithms. In the case of PSO, the performance enhancement will be done by changing the parameters of Eqs. 2 and 3.

Several studies published in the past present considerable improvement in the performance of PSO by changing parameters in the objective function and equations for velocity/position or by introducing new relevant parameters in the algorithm (Imran et al. 2013; Strąk et al. 2019). The addition of V_{max} and specific boundary conditions may also help to reduce error. On the other side, the time taken by GA is also required to be reduced without damaging its performance. The main considerations that can be used for achieving a balanced performance in the future are population size, the number of decision variables, computer hardware configuration, the complexity of fitness function, and moderate levels of threshold (Trujillo et al. 2015; Vlašić et al. 2019). Moreover, most of the existing studies in the literature so far are either limited to laboratory experiments or do not provide efficient results for real-time implementations.

On the one hand, the authors are planning to propose a stand-alone monitoring and prediction system that could be installed in real-time scenarios in rural and urban settings. The future system will be linked to a mobile app that can provide continuous updates about pollutant concentrations in the indoor environment while generating instant alerts for critical threshold levels. On the other hand, the main consideration for the development of standalone monitoring and forecasting system is to ensure cost-effective design, lesser resource utilization, and responsiveness for real-time applications.

5 Conclusion

IAP is a critical area of concern for urban and rural areas. The impact is equally high in developed and developing countries due to the extensive range of population who spends 80 to 90% of their time indoors. FIS is a widely accepted, soft computing approach for forecasting purposes. Besides, the fuzzy modeling techniques can analyze

and interpret any kind of information (logical, linguistic, and numeric) with self-tuning, self-organizing, and self-learning abilities. We proposed a novel application for optimizing FIS performance with PSO and GA to achieve improved forecasting accuracy for real-time scenarios. To the best of our knowledge, none of the existing studies have tested the proposed method for PM_{10} prediction. The model focuses on important IAQ parameters (PM, CO2, VOC, temperature, and humidity) that are closely associated with human health and well-being. The results show that the FIS-GA model with normalized dataset presents a better performance in terms of RMSE reduction (RMSE = 0.998). Moreover, the optimized FIS-PSO model with normalized dataset shows that this technique does not just reduce the output error (RMSE = 1.0746). Instead, the computational cost is also reduced due to the lesser number of algorithm parameters. This study executes correlation-based feature selection to improve model performance so that an ideal response for forecasting can be obtained.

Nevertheless, this model makes use of a limited number of parameters with six-month data consisting of 17,280 samples. Moreover, the prediction is performed only for one variable (PM_{10}). Other than this, the optimized models are required to be balanced in terms of accuracy and time taken to produce the best results. In the future, we plan to design a forecasting model that performers well with an extensive range of seasonal data and can predict the concentration of multiple pollutants for the building environment. Furthermore, we aim to design a standalone system in the future that can be installed in the indoor areas to get instant updates about IAQ conditions. The future system will provide instant alerts about pollutant concentration and threshold levels via a mobile app. The main goal is to design a cost-effective and efficient monitoring and forecasting system that could be installed in rural and urban areas to prevent critical consequences associated with IAP.

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Declarations

Conflict of interest Authors declare no conflicts of interest.

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